TRACE: Efficient Transfer Learning for Neural Whole-Body Control Across Humanoid Embodiments

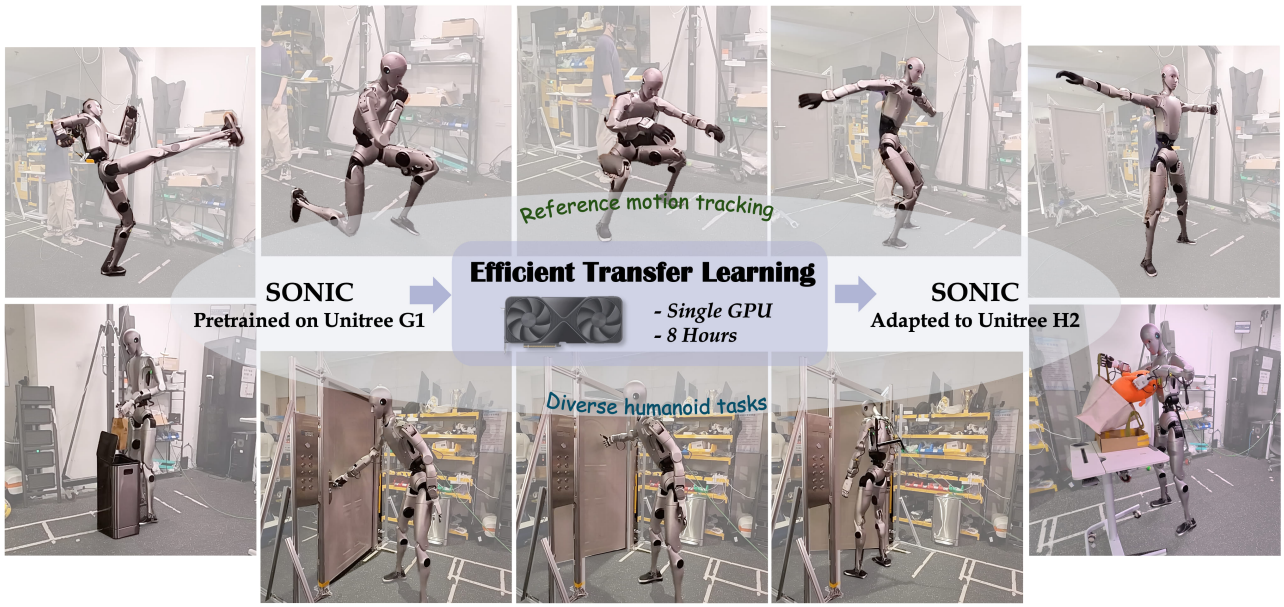


Fig. 1. TRACE enables efficient cross-embodiment transfer of humanoid behavior foundation models. We adapt the pretrained SONIC controller from the compact Unitree G1 to the full-scale Unitree H2 in approximately eight hours of training on a single GPU. The resulting controller supports whole-body motion tracking and diverse downstream tasks.

***Abstract*— Humanoid behavior foundation models (BFMs) enable diverse whole-body control, but adapting a pretrained controller to a new embodiment remains challenging because of kinematic and dynamic mismatch. We present TRACE, a framework for efficient cross-embodiment adaptation that preserves pretrained motion representations while specializing feedback control to the target robot. TRACE first injects kinematic priors into input and output projection layers and aligns nominal joint dynamics to reduce embodiment mismatch and support stable fine-tuning. It then combines encoder distillation with low-rank reinforcement learning on a compact, quality-screened motion set, using a two-phase curriculum to establish stable tracking before refining difficult motions. We apply TRACE to transfer SONIC from Unitree G1 to Unitree H2 using 10 million motion frames and approximately eight hours of training on a single RTX PRO 6000 GPU, at less than 0.1% of the GPU-hour cost of training SONIC from scratch.* The adapted controller achieves 96.08% success on 67,953 evaluation motions under a height-scaled tracking criterion. Real-robot demonstrations span dynamic motion tracking and whole-body teleoperation, with collected demonstrations supporting downstream vision–language–action policy post-training. These results demonstrate practical reuse of a pretrained humanoid controller across robots with different body scales and dynamics.**

*SONIC's reported training cost of 21,000 GPU-hours [2] serves as the reference. TRACE's cost covers only adaptation from the released SONIC checkpoint, excluding source pretraining. GPU-hours are not normalized across hardware.

I. Introduction

Humanoid behavior foundation models (BFMs) aim to learn a single whole-body controller for diverse behaviors rather than behavior-specific policies [1]. Training on large motion datasets with a shared tracking objective makes this approach scalable. Reference motions also provide a compact representation and command interface for higher-level models to specify whole-body behavior. SONIC exemplifies this paradigm: trained on over 100 million motion frames, a single Unitree G1 policy supports tracking unseen motions, VR teleoperation, and vision–language–action (VLA) control [2].

Although a BFM trained for a single embodiment can generalize across behaviors, cross-embodiment transfer requires addressing two principal mismatches. First, kinematic mismatch stems from differences in the number and arrangement of actuators and sensors, as well as in link geometry, altering the dimensionality and physical semantics of the BFM's observation and action spaces. Second, dynamics mismatch may persist even after kinematic correspondence is established. Differences in link masses and other inertial properties, actuator dynamics, and contact dynamics may induce different state transitions under identical commands and feedback, leading to different closed-loop responses that

can affect stability. Neither mismatch is readily resolved by modifying only the BFM's inputs and outputs. Yet training a dedicated BFM for each embodiment is prohibitively expensive, particularly as robotic platforms evolve rapidly.

Recent multi-embodiment approaches jointly train shared controllers across multiple humanoid embodiments or randomized morphologies, often using unified observation–action interfaces and embodiment-aware architectures [3]–[5]. Their goal is zero-shot generalization or few-shot adaptation to new embodiments [3], [4]. However, broader embodiment coverage increases training cost, while control performance on out-of-distribution embodiments remains uncertain [3], [4]. Shared policies may also fall short of embodiment-specific policies in control accuracy and stability [4], [5]. Recent work also enables efficient transfer of pretrained BFMs by combining joint-level kinematic alignment with low-rank adaptation of dynamics-sensitive modules [26]. However, this formulation treats the post-alignment gap as dynamical, without an explicit objective for aligning motion latents across embodiments. It also retains the source motion-sampling and optimization settings, leaving transfer-specific data selection and staged control refinement outside its scope.

To this end, we introduce TRACE, an efficient **TR**ansfer learning framework **AC**ross humanoid **E**mbodiments for BFMs. TRACE reuses pretrained motion representations while specializing feedback control for the target embodiment. We first inject kinematic priors into the BFM's input and output projection layers for kinematic pre-alignment, and further reduce the initial embodiment mismatch through joint-command normalization and dynamic-response alignment. We also introduce a targeted dataset construction strategy that selects a compact transfer set from quality-screened motions, balancing motion categories and promoting dispersion in the command feature space. Using this set, we separate representation alignment from control optimization during transfer training: encoder distillation anchors target motion encodings to the pretrained latent space, while reinforcement learning adapts the control decoder to target dynamics, both through low-rank updates. To accelerate convergence and improve control quality, we organize this training into two phases: first establishing stable tracking across the selected motions, then emphasizing difficult motions to improve tracking and contact accuracy.

We use TRACE to transfer the pretrained SONIC controller from Unitree G1 to Unitree H2. The two embodiments differ in body scale, mass distribution, joint geometry, and actuation. The transfer training completes in approximately eight hours on a single RTX PRO 6000 GPU, at a fraction of SONIC's 21k GPU-hour pretraining cost [2] and with improved transfer efficiency over prior work [26]. Quantitative evaluation on a common motion-tracking benchmark validates the control quality achieved after transfer, while ablation studies quantify the contributions of key components. On the physical Unitree H2, we further demonstrate demanding motion tracking and collect whole-body demonstrations through VR teleoperation. These demonstrations are then used to post-train a VLA model for autonomous task execution. These results show that TRACE enables efficient cross-embodiment BFM transfer and provides a reusable controller for real-world tasks and downstream policy learning.

The main contributions of this work are threefold:

- We present TRACE, an efficient cross-embodiment transfer learning framework for BFMs that reuses pretrained motion representations while specializing feedback control for a new humanoid. Its core design combines kinematic and dynamic-response pre-alignment with two-phase low-rank training that decouples reference-encoder distillation from control optimization.
- We introduce a targeted motion-selection strategy that balances motion categories and promotes dispersion in the command feature space after quality screening. The resulting compact transfer set improves motion-tracking success under a fixed data budget.
- We achieve low-cost transfer of SONIC from Unitree G1 to the substantially different Unitree H2, with 96.08% tracking success on the PHUMA benchmark. On the physical H2, the transferred controller supports demanding motion tracking and VR-based whole-body demonstration collection, with the collected data enabling VLA post-training for autonomous tasks.

II. Related Work

a) Humanoid Behavior Foundation Model: Humanoid BFMs build on reference-motion tracking in physics-based character control. DeepMimic learned robust, physically plausible policies with dense imitation rewards over prescribed clips; later work extended this to diverse motions, dynamic and balance-intensive skills, and adaptation to real humanoids [6]–[9]. Early generalists used broader motion libraries or unified command interfaces within a single policy for universal tracking, teleoperation, and multimode whole-body control [10]–[12], extending beyond individual references or small skill sets without jointly scaling motion data, policy capacity, and distributed compute. Complementary unsupervised reinforcement learning prioritizes stability and safety over fine-grained tracking [13]. SONIC established foundation-scale reference-conditioned control by jointly scaling data, policy capacity, and compute; trained on over 100 million motion frames, it generalizes to unseen motions [2]. Later work explores data and model scaling with billion-frame corpora and causal Transformers [14], and capacity scaling via conditional routing in sparsely activated mixture-of-experts Transformers [15]. Existing BFMs remain coupled to their training embodiment's state–action interface and closed-loop dynamics.

b) Multi-Embodiment Generalist Control: Multi-embodiment control assumes shared control-relevant physical regularities across robots with different kinematics and dynamics. Early shared policies encoded embodiment and pooled cross-design experience, enabling transfer to

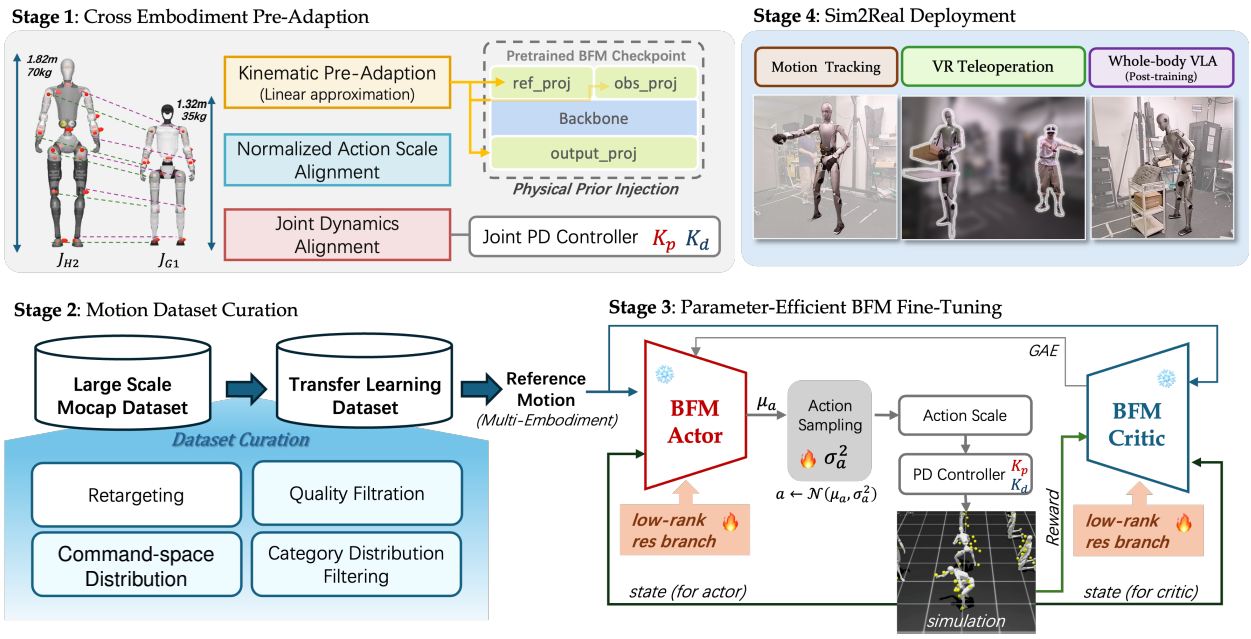


Fig. 2. Overview of TRACE. *Stage 1 (top left):* projection layers are inserted at the reference, observation, and action interfaces to inject kinematic priors; action scales are calibrated and nominal joint responses are aligned. *Stage 2 (bottom left):* retargeting, quality screening, category balancing, and command-space selection form a compact motion dataset. *Stage 3 (bottom right):* with base weights frozen within each training phase, low-rank residual branches adapt the target Robot reference encoder through latent and token distillation, and the Robot Control Decoder through PPO in target simulation (Fig. 3). *Stage 4 (top right):* the transferred controller supports diverse robot tasks, including reference-motion tracking, VR teleoperation, and downstream VLA-based loco-manipulation.

unseen structures within a morphology space [16]. Legged-robot methods combine morphology-agnostic diffusion with shared residual adaptation [17], or large-scale training on procedurally generated bodies with long-context test-time adaptation [18]. Recent humanoid systems combine unified state–action semantics, physically consistent morphology variation, embodiment- and topology-aware policies, and generalist–specialist refinement for zero- or few-shot high-dimensional whole-body control and efficient target specialization [3]–[5]. These methods amortize learning across embodiments, but training cost and coverage depend on the training distribution's breadth and physical validity. Joint tokenization of embodiment, state, and action also supports robot co-design and cross-embodiment control [19], but scalability to humanoid BFM pretraining remains unverified.

c) Transfer Learning for Humanoid BFMs: BFM transfer reuses pretrained motion priors rather than relearning control. Same-embodiment approaches typically add residual or delta policies to pretrained trackers to adapt to new motion and task distributions or improve execution accuracy [20]–[23]. With fixed kinematics and observation–action interfaces, they do not directly address embodiment transfer. Earlier cross-embodiment studies progressively optimized policies along continuous morphology paths to narrow embodiment gaps, but required repeated training and did not address parameter-efficient transfer of large-scale humanoid BFMs [24], [25]. Recent work first validated parameter-efficient fine-tuning (PEFT) for cross-embodiment humanoid BFM transfer [26]: kinematic input/output alignment and LoRA [27] fine-tuning of dynamics-related modules enable low-cost transfer to multiple embodiments, indicating that low-rank updates effectively capture this adaptation. However, residual kinematic errors after joint-level alignment shift the command distribution, while overemphasizing difficult motions early in fine-tuning erodes pretrained priors; both slow convergence and reduce final tracking accuracy.

III. Methodology

Figure 2 presents the four-stage TRACE framework. Cross-embodiment pre-adaptation (Stage 1, Sec. III-B) injects kinematic priors into reference, observation, and action projections and aligns action scales and nominal joint responses. Motion dataset curation (Stage 2, Sec. III-C) combines retargeting and quality screening with category balancing and command-space selection. Parameter-efficient fine-tuning (Stage 3, Sec. III-D) distills command latents and tokens from the frozen source Robot reference encoder into its target counterpart, while PPO adapts the control decoder to target dynamics. The transferred controller is then deployed on the target robot (Stage 4), supporting reference-motion tracking, VR teleoperation, and downstream VLA-based loco-manipulation through a shared command interface.

A. Problem Formulation

BFMs are pretrained by large-scale motion tracking, formulated as a discounted MDP. For embodiment $r \in \{s, t\}$,

the policy $\pi_r(u_r \mid o_r, g_r)$ receives proprioception o_r and a motion reference g_r, and outputs normalized joint commands u_r. These define position targets for a PD controller; rewards measure motion-tracking quality, and transitions follow the embodiment's dynamics. We primarily focus on transferring SONIC-based BFMs from a half-scale source humanoid s to a full-scale target humanoid t. Our goal is to accommodate differences in joint layouts and dynamics using limited target-side training while retaining the pretrained motion representation.

B. Cross-Embodiment Pre-Adaptation

We align joint coordinates, action units, and nominal servo responses to reduce initial tracking errors and stabilize early optimization while preserving the source BFM's internal representation.

a) Kinematic prior injection: To accommodate different joint layouts, we retain the pretrained model's source joint-coordinate interface and inject kinematic priors into its reference, proprioceptive, and action projection layers. We estimate the required linear maps from paired retargeted poses, regularized by a Jacobian-based prior.

Let $\Delta q_r = q_r - q_r^0$ denote joint offsets from nominal pose q_r^0, and let $Q_r \in \mathbb{R}^{n_r \times N}$ collect N paired joint-offset samples for n_r joints. We construct the local prior $M_0 = (J_t^0)^\dagger D_{s\to t} J_s^0$ from limb-endpoint Jacobians. Here J_r^0 stacks the endpoint translational Jacobians at the nominal pose in aligned limb-root frames, $D_{s\to t}$ scales each endpoint's three coordinates by the corresponding target-to-source nominal root-to-endpoint length ratio, and $\dagger$ denotes the Moore–Penrose pseudoinverse. We fit the forward map by

$$M_{s\to t}^\star = \arg\min_M \frac{1}{N}\|MQ_s - Q_t\|_F^2 + \lambda_J \|M - M_0\|_F^2, \quad (1)$$

where $\lambda_J > 0$ controls the prior strength. The data term fits joint coordination across poses, while the prior anchors directions poorly covered by the data.

The same constant map is used for joint offsets and velocities: $\Delta q_t \approx M_{s\to t}^\star \Delta q_s$ and $\dot{q}_t \approx M_{s\to t}^\star \dot{q}_s$. Exchanging s, t in the fit and prior gives a separately fitted reverse map. We embed it in the full reference and proprioceptive input maps $x_s = P_i x_t + c_i$, $i \in \{\mathrm{ref}, \mathrm{obs}\}$, leaving shared-coordinate channels unchanged. Here c_i contains nominal-pose offsets, and P_{act} denotes the normalized source-to-target action map in (3). Before fine-tuning, source projection weights W^s and biases b^s become target parameters W^t, b^t:

$$\begin{aligned} W_i^t &= W_i^s P_i, & b_i^t &= b_i^s + W_i^s c_i, \\ W_{\mathrm{act}}^t &= P_{\mathrm{act}} W_{\mathrm{act}}^s, & b_{\mathrm{act}}^t &= P_{\mathrm{act}} b_{\mathrm{act}}^s. \end{aligned} \quad (2)$$

Input projections therefore use right multiplication, whereas the action projection uses left multiplication.

b) Action-scale alignment: The kinematic map relates physical joint offsets, whereas the decoder outputs normalized actions. Let $S_r = \mathrm{diag}(s_r)$, where $s_{r,j} > 0$ is the joint-angle offset per unit action. The target action $u_t = P_{\mathrm{act}} u_s$ and its PD position target are defined by

$$\begin{aligned} P_{\mathrm{act}} &= S_t^{-1} M_{s\to t}^\star S_s, \\ q_t^{\mathrm{cmd}} &= q_t^0 + S_t \,\mathrm{clip}_{\mathcal{A}_t}(u_t), \end{aligned} \quad (3)$$

where clipping enforces the target normalized-action range $\mathcal{A}_t$.

c) Joint dynamics alignment: To reduce actuator-response mismatch, we initialize target PD gains using the source controller's common nominal natural frequency ω_n and damping ratio ζ. Under a decoupled actuator model,

$$K_{p,t,j}^{(0)} = I_{t,j}\omega_n^2, \qquad K_{d,t,j}^{(0)} = 2\zeta\omega_n I_{t,j} - d_{t,j}, \quad (4)$$

where $I_{t,j} > 0$ and $d_{t,j} \geq 0$ denote joint-side reflected actuator inertia and passive damping at the nominal pose, with $K_{d,t,j}^{(0)} > 0$. This actuator-level initialization excludes link and contact dynamics, which are addressed by policy fine-tuning.

C. Motion Dataset Curation

Efficient transfer requires reliable motion references within a limited data budget. We combine quality screening with category balancing and command-space distribution shaping to construct the transfer set.

a) Retargeting and quality screening: We retarget the motion corpus to both embodiments, obtaining paired references for kinematic calibration and policy transfer. Quality screening rejects discontinuities, sustained joint-position or velocity-limit violations, unintended collisions, and implausible clip-level acceleration or jerk. Contact checks detect excessive penetration, support-point sliding, and missing intended support while preserving annotated object contacts.

b) Category balancing: To avoid overrepresenting common motions, we divide a fixed frame budget equally across ten metadata-derived categories (Fig. 6(b)) in a fixed, quality-screened candidate pool. If a category has fewer frames than its quota, we retain all its clips. We redistribute the unused budget equally among categories with more frames than their quotas.

c) Command-space distribution: To promote command-space dispersion within each category, we summarize frozen SONIC reference-encoder window features by their temporal means and standard deviations, forming a 128-dimensional clip descriptor. Let E_i be the mean squared component value of its standardized descriptor and $\widetilde{E}_i$ its counterpart after regularized covariance whitening. Standardization parameters and covariance are estimated from the full valid source pool. The clip score is

$$s_i = \tfrac{1}{2}\log\max(E_i, \epsilon) + \tfrac{1}{2}\log\max(\widetilde{E}_i, \epsilon), \quad (5)$$

where $\epsilon = 10^{-12}$ avoids taking the logarithm of zero. E_i measures departure from the pool center in standardized coordinates, while $\widetilde{E}_i$ additionally accounts for feature correlations. Their logarithms are equally weighted, and the resulting score guides selection toward larger combined deviations within each category's frame budget. Paired source–target references remain synchronized for distillation and tracking.

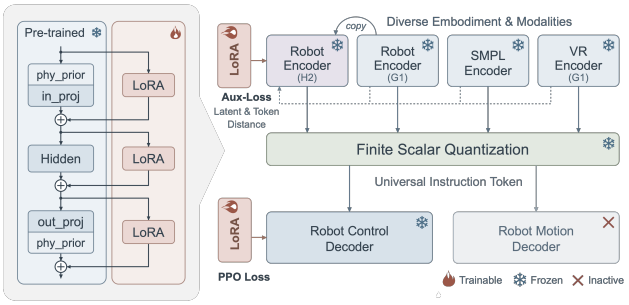


Fig. 3. SONIC adaptation with kinematic priors and LoRA. Left: priors are folded into projection weights, and LoRA augments frozen layers. Right: the target Robot encoder uses latent and token distillation; the control decoder uses PPO. Source encoders and FSQ remain frozen.

D. Parameter-Efficient Fine-Tuning

SONIC encodes references as Universal Tokens and combines them with proprioception in a Robot Control Decoder. We adapt reference encoding and feedback control separately to address residual representation and dynamics mismatch.

a) Model architecture and LoRA insertion: As shown in Fig. 3, we retain the source reference motion encoders (Robot, Human, and Hybrid) as frozen branches. We initialize a single target Robot reference encoder from its source counterpart with the pre-adapted interface. All branches share the fixed finite scalar quantization (FSQ) operator.

LoRA [27] augments linear layers in the target Robot reference encoder and Robot Control Decoder:

$$W_{\text{eff}} = W_b + \frac{\alpha}{r_{\text{L}}} BA, \tag{6}$$

where W_b is frozen within each training phase, α is the scale, and trainable factors A, B have inner dimension r_{L}. LoRA branches in both the target Robot reference encoder and control decoder use Kaiming-uniform initialization for A and $B = 0$, preserving their pre-adapted mappings at insertion. The critic uses LoRA in every linear layer except its trainable final scalar projection. The Robot Motion Decoder remains frozen and excluded from training because target-motion reconstruction is not required for feedback control.

b) Command latent alignment: Preserving command-token semantics is a transfer requirement for reusing multitask interfaces, including VR teleoperation, motion generators, and planners. Unconstrained end-to-end RL risks semantic drift, whereas freezing the reference encoder leaves residual kinematic-mapping errors that can shift token distributions after nonlinear encoding and FSQ. We therefore fine-tune only the target Robot reference encoder against its frozen source counterpart to maintain command compatibility and support stable, data-efficient controller fine-tuning. Let $\mathbf{z}_s$ and $\mathbf{z}_t$ denote their pre-FSQ command latents, and $\mathbf{c}_r = \mathcal{Q}(\mathbf{z}_r)$, $r \in \{s, t\}$, their quantized token vectors under the fixed FSQ operator $\mathcal{Q}$. We jointly align both representations by minimizing

$$\mathcal{L}_{\text{align}} = \ell_{\text{SL1}}(\mathbf{z}_t, \mathbf{z}_s) + \ell_{\text{SL1}}(\mathbf{c}_t, \mathbf{c}_s)\,, \tag{7}$$

where the same smooth-L_1 loss ℓ_{SL1} measures both latent and token distances.

c) Controller fine-tuning with PPO: During rollouts, we sample the retained source reference branches and the adapted target Robot branch according to prescribed mixture proportions and use their quantized tokens as commands. Each token remains paired with the corresponding target trajectory for tracking-reward evaluation. PPO updates the Robot Control Decoder's LoRA parameters and the action-sampling variance, with gradients stopped at the token input; the critic receives the value loss.

d) Two-phase transfer learning curriculum: Premature concentration of the training distribution on difficult motions can disrupt the BFM's pretrained motion knowledge. We therefore adopt a two-phase training curriculum within Stage 3. First, we sample from the balanced transfer-set distribution and strengthen existing global-position and end-effector tracking rewards to establish stable target tracking. At the phase transition, we merge LoRA updates into the base weights and reinitialize zero-output branches. Second, we adaptively prioritize motion segments with frequent failures or large tracking errors while retaining baseline sampling for coverage. We further strengthen end-effector tracking rewards and introduce terrain-interaction and robustness regularization to improve precision and disturbance rejection. Final LoRA updates are merged for deployment.

IV. Experiments

We evaluate whether TRACE transfers broad tracking capability under a limited adaptation budget and supports downstream use. Embodiment differences and the availability of pretrained target policies and compatible implementations limit direct comparisons with prior transfer methods. We therefore emphasize training baselines and component ablations on the target robot, using the source controller as a reference for tracking quality in simulation. On hardware, where global tracking errors are difficult to measure reliably, we assess challenging whole-body motions and downstream task performance.

A. Experimental Setup

a) Training configuration: We adapt the pretrained SONIC controller [2] from Unitree G1 to Unitree H2 in simulation, following Sec. III. The adaptation set contains 10M motion frames selected from BONES-SEED [28]. Unless otherwise specified, LoRA modules use rank $r_{\text{L}} = 32$. Training uses 12,288 parallel environments, a rollout horizon of 24 steps, and a batch size of 1,536.

b) Baselines: We compare TRACE with *Direct Fine-tune* (full-parameter fine-tuning) and *Scratch* (training without pretraining). Convergence is evaluated over the first 3k iterations under a common motion-sampling distribution, with difficulty-based oversampling disabled. We report completion rate, measured as the episode-timeout fraction, and mean training reward.

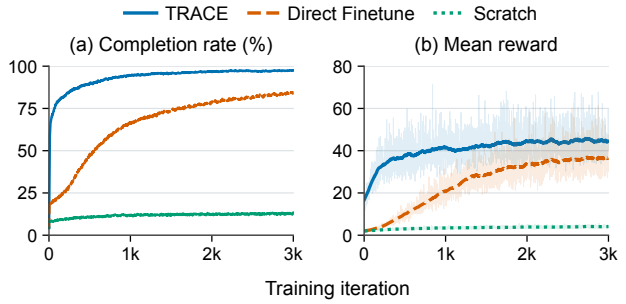


Fig. 4. Convergence under a common motion-sampling distribution (difficulty-based oversampling disabled), with one run per method: (a) completion rate; (b) mean reward. TRACE converges within a small iteration budget; Direct Finetune requires substantially more training. Scratch fails to obtain a usable tracking policy within the 3k-iteration budget.

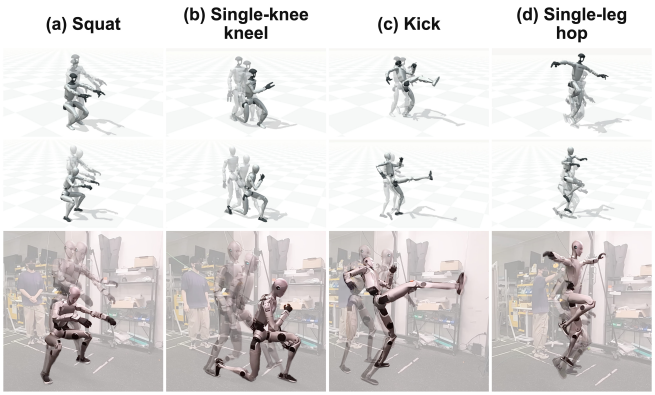


Fig. 5. Motion tracking across embodiments and deployment settings. From top to bottom, rows show Unitree G1 simulation, Unitree H2 simulation, and H2 hardware execution. Columns show (a) squatting, (b) single-knee kneeling, (c) kicking, and (d) single-leg hopping.

TABLE I

Results on the PHUMA benchmark. SR uses body-size-adjusted thresholds: 0.250 m (G1) and 0.3447 m (H2). Errors are evaluated only on each policy's successful motions: E_{loc} in mm, E_{vel} in mm/frame, and E_{acc} in mm/frame^2, at 50 Hz.

Method	Robot	SR (%)↑	E_{loc} (mm)↓	E_{vel} ↓	E_{acc} ↓
SONIC (Pre-trained)	Unitree G1	97.59	27.81	5.157	1.390
TRACE (full)	Unitree H2	96.08	55.81	7.536	2.394
w/o-Pre-Adaption	Unitree H2	87.29	72.88	8.033	2.474
w/o-Data-Strategy	Unitree H2	93.80	54.01	7.853	2.558
w/o-Two-Phase-FT	Unitree H2	95.64	62.47	8.115	2.553

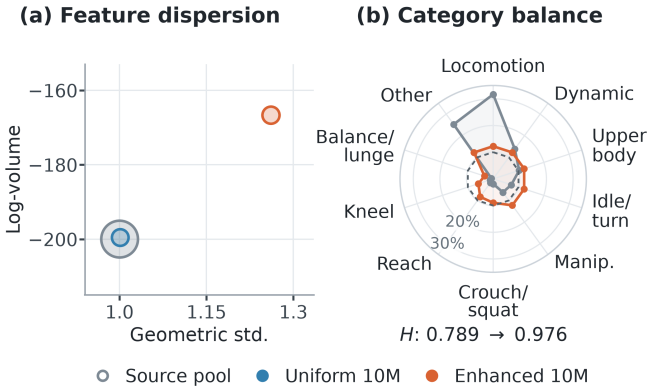


Fig. 6. Data selection statistics. (a) Geometric mean of feature standard deviations and regularized covariance log-volume of frozen SONIC descriptors under shared standardization. Coordinates average 20 draws of 5,000 clips; bubble areas indicate clip counts. (b) Category frame shares; the dashed polygon marks equal shares (10%). H denotes normalized category entropy.

c) Evaluation: We evaluate motion tracking in simulation on the PHUMA benchmark [29], using 67,953 motions excluded from H2 adaptation training. We evaluate the late-training TRACE/H2 policy and re-evaluate the released SONIC/G1 policy on the same motion set, with references retargeted to each robot. We report clip success rate (SR), root-relative mean body-position error E_{loc} (MPJPE-L), velocity error E_{vel}, and acceleration error E_{acc}.

B. Transfer Efficiency and Control Quality

a) Convergence within a limited budget: TRACE converges within a small number of training iterations, reaching stable completion and reward levels (Fig. 4). Direct Finetune converges substantially more slowly, approaching TRACE's mean reward only later in training. Scratch remains at low completion and reward levels and does not produce a usable tracker within the 3k-iteration budget.

b) Motion Tracking Capability: TRACE retains a broad tracking repertoire after transfer, completing 96.08% of the evaluation motions, within 1.51 percentage points of SONIC/G1 (Table I). Local position error is 55.81 mm on H2 versus 27.81 mm on G1, with higher temporal errors on H2. SR uses the height-based success criterion in [2]: the absolute world-height errors of the root, wrists, and ankles must remain within the threshold at every evaluated frame of a clip. Scaling the G1 threshold by the whole-body height ratio $1.820/1.320$ preserves the same error tolerance relative to robot height.

Figure 5 shows the adapted controller executing squats, single-knee kneels, kicks, and single-leg hops in simulation and on H2 hardware. These motions span large posture changes, single-leg support, and dynamic limb coordination.

C. Ablation of the Adaptation Design

a) Pre-adaptation: Kinematic mismatch and learned control corrections interact during BFM transfer learning. We therefore assess pre-adaptation as a complete initialization, encompassing kinematic prior injection, action-scale alignment, and nominal joint-response alignment. This comparison evaluates their combined effect without independently resolving each component's contribution. Omitting pre-adaptation before target-robot training reduces SR from 96.08% to 87.29% and increases E_{loc} from 55.81 to 72.88 mm (Table I), producing the largest SR drop among the three component ablations. These results support jointly

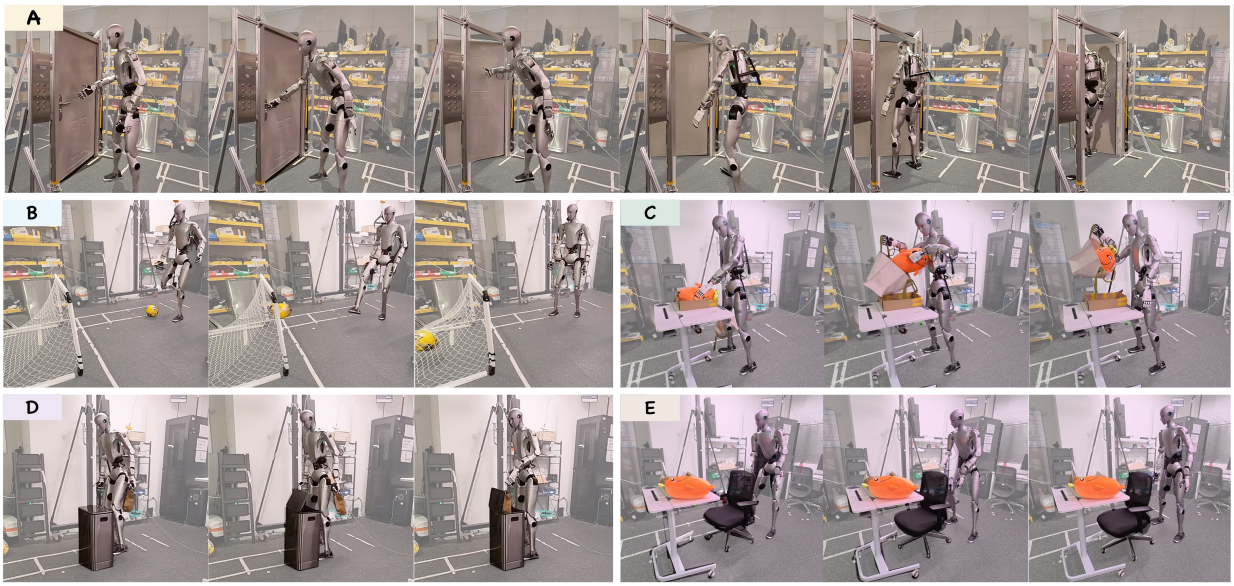


Fig. 7. Whole-body teleoperation and downstream applications on H2: (A) door opening and passage, (B) soccer, (C) toy packing, (D) trash disposal, and (E) chair repositioning. Keyframes progress left to right, showing execution across objects and contact conditions.

TABLE II

Final H2 SONIC transfer training statistics, with simulation rollouts using the pretrained G1 checkpoint under uniform sampling as a reference. Position errors: cm; joint-angle error: rad.

Metric	Single-phase	Two-phase	Δ (%)	G1 sim rollout
Training reward ↑	62.57	70.04	+11.9	—
Root-position error ↓	42.45	31.34	−26.2	18.62
Global foot error ↓	42.94	31.81	−25.9	19.04
Local hand error ↓	7.09	6.40	−9.7	6.31
Joint-angle error ↓	0.12555	0.11957	−4.8	0.1146
Critic value loss	0.28871	0.20108	−30.4	—

TABLE III

VLA task success rates on the physical H2 robot.

Task	Type	Trials	Success (%)
Object relocation (toy)	Single-arm	20	90
Object relocation (cube)	Bimanual	20	75
Object pickup (pillow)	Whole-body	20	80
Open door	Whole-body	20	65
Open trash can	Whole-body	20	65

aligning robot interfaces and nominal responses before fine-tuning to improve the quality of transferred whole-body control.

b) Data strategy: The data strategy balances category frame budgets and ranks clips within each category using frozen SONIC descriptors (Sec. III-C). We first examine the resulting distributions by comparing uniform and enhanced 10M-frame subsets from the same candidate pool (Fig. 6). Uniform sampling remains close to the source pool in feature statistics, whereas enhanced selection increases the geometric mean of feature standard deviations from 1.001 to 1.261 and yields a higher covariance log-volume. It also increases the frame shares of underrepresented crouching, reaching, kneeling, and balance motions, raising normalized category entropy from 0.789 in the source pool to 0.976 in the selected set.

We then assess tracking behavior by comparing full TRACE with the variant without the data strategy (Table I). Removing the strategy reduces SR from 96.08% to 93.80%, a drop of 2.28 percentage points, and increases both velocity and acceleration errors. In rollouts, the ablated policy tracks common motions well but fails on some highly dynamic motions that full TRACE completes. Its slightly lower E_{loc} (54.01 versus 55.81 mm) is averaged over a smaller successful-motion set that excludes these failures; full TRACE's average includes additional challenging executions. The SR gain and rollout observations support the role of data selection in broadening the adapted controller's motion repertoire.

c) Two-phase transfer learning curriculum: We compare the two-phase curriculum with a single-phase baseline at the training endpoint (Table II). The curriculum starts with balanced motion sampling, then merges the LoRA updates, reinitializes the adapters, and introduces difficult-motion resampling. The baseline uses the second training phase's reward terms, weights, and other hyperparameters throughout, including difficult-motion resampling from the outset. For matched cumulative LoRA rank budgets, the baseline uses rank $2r_{\mathrm{L}} = 64$, versus $r_{\mathrm{L}} = 32$ per training phase.

Two-phase fine-tuning increases reward by 11.9% and reduces root-position and global foot errors by 26.2% and 25.9%, with lower local hand and joint-angle errors (Table II). The H2/G1 global-error ratios are broadly in line

with their body-size ratio; G1 rollouts exclude difficult-motion resampling, favoring lower errors. Removing the curriculum also lowers benchmark SR to 95.64% and raises E_{loc} to 62.47 mm (Table I). In our runs, early difficult-motion resampling kept policy standard deviation elevated, supporting stable, broad tracking before emphasizing difficult motions.

D. Whole-Body Control and Downstream Applications

a) Whole-body interaction and teleoperation: Figure 7 shows H2 performing door opening and passage, toy packing, trash disposal, chair repositioning, and soccer. The transferred controller coordinates body motion, manipulation, and foot contact, while the teleoperation interface supports task execution and demonstration collection.

b) Integration with VLA policies: Motion generation and VLA share the same H2 low-level BFM weights and motion interface. We post-trained GR00T-N1.7 [30] on selected tasks. Across five VLA tasks with 20 trials each, success rates span 65%–90% (Table III), covering single-arm, bimanual, and whole-body execution, including trash-can opening with foot contact.

V. Conclusion

TRACE demonstrates that a pretrained humanoid BFM can be efficiently adapted to a new embodiment while retaining a shared motion interface for tracking and downstream tasks. The framework combines kinematic and nominal joint-response pre-alignment with a compact motion set selected for quality, category balance, and command-space dispersion. It decouples reference-encoder distillation from feedback-control optimization within a two-phase low-rank curriculum, establishing stable tracking before improving accuracy on difficult motions. TRACE transfers SONIC from Unitree G1 to H2 in approximately eight hours on a single GPU while retaining a broad motion-tracking repertoire. On physical H2, the same adapted controller supports dynamic motion tracking, teleoperated demonstration collection, and autonomous VLA tasks.

Our evaluation is limited to one embodiment pair and one pretrained BFM. Future work will extend validation to more embodiments and open-source BFMs and investigate efficient adaptation without a reusable critic. For example, multi-expert distillation may provide no unified critic, whereas changes in training paradigm may make existing value estimates unsuitable for the new objective. We will also explore incorporating RGB, depth, and tactile observations during transfer to develop perceptive BFMs.

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